Software Requirements Specification (SRS) Template

Project: Online Quiz System
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Revision history

Version	Date	Author	Change summary	Approval
1.0	28-09-2026	Debangana, Chinmay, Satvik, Sunil	SRS prepared for Online Quiz System	

Approvals

Role	Name	Signature / Email	Date
Course Coordinator			

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1. Introduction

1.1 Purpose

This document describes the requirements of the Online Quiz System. It explains what the system should do, who will use it, and the basic conditions the software must satisfy.

1.1.Scope

The system is a console-based quiz application developed in C/C++. It allows users to answer multiple-choice questions, checks answers, calculates the score, and displays the final result. An administrator can manage quiz questions.

1.2.Audience

This document is intended for students, developers, testers, instructors, and anyone involved in reviewing or maintaining the project.

1.3.Definitions

MCQ – Multiple Choice Question; UI – User Interface; Admin – Administrator; Quiz – A set of questions answered by the user; Score – Number of correctly answered questions.

2. Overall description

2.1.Product perspective

The Online Quiz System is a standalone console application. It takes input from the user through the keyboard, processes quiz questions and answers, and displays the result on the console.

2.2.Major product functions

- Start a quiz
- Display multiple-choice questions
- Accept answers
- Validate answers
- Calculate the score
- Display the final result
- Allow an administrator to add questions
- Allow an administrator to modify questions
- Allow an administrator to delete questions
- Validate user input

2.3.User roles and characteristics (expanded)

- User: selects answers and views the final score. Basic computer knowledge is sufficient.
- Administrator: manages quiz questions and requires basic knowledge of the system.

2.4.Operating environment

The system runs as a C/C++ console application on a computer with a suitable C/C++ compiler. It can run on common desktop operating systems such as Windows or Linux.

2.5.Constraints

The first version is console-based and depends on keyboard input. The quiz data must be stored in a format that can be read and updated by the application. The system does not require a web browser or internet connection for basic quiz operation.

3. External interface requirements

3.1.User interfaces

The system uses a console interface. Questions, options, instructions, error messages, and scores are displayed as text. The user enters an option such as A, B, C, or D.

3.2.Hardware interfaces

No special hardware is required. The system uses the computer keyboard for input and the display for output.

3.3.Software interfaces

The application uses standard C/C++ libraries for input, output, strings, and data handling. A local file may be used to store quiz questions.

3.4.Communications

No network communication is required for the basic version of the system.

4. System features

Each requirement below includes acceptance criteria and a reference test case. IDs follow OQ-F-###.

4.1.Quiz Operation

Description: Define the main quiz operations including starting the quiz, displaying questions, accepting answers, checking answers, and calculating the score.

Req ID	Requirement	Type	Priority	Source/ Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
OQ-F-001	The system shall start a new quiz when the user selects the start option.	Functional	High	User	Quiz starts and the first question is displayed.	

OQ-F-002	The system shall display one multiple-choice question at a time.	Functional	High	User	A question with its available options is displayed.	
OQ-F-003	The system shall display four answer options for each standard MCQ.	Functional	High	User	Options A, B, C and D are shown.	
OQ-F-004	The system shall accept the user's selected answer.	Functional	High	User	A valid option is accepted as the answer.	
OQ-F-005	The system shall reject an invalid answer option and ask the user to enter a valid option.	Functional	Medium	User	Invalid input produces an error message and the question remains active.	
OQ-F-006	The system shall compare the selected answer with the correct answer.	Functional	High	System	The answer is marked correct or incorrect.	
OQ-F-007	The system shall increase the score when the selected	Functional	High	User	A correct answer increases	

	answer is correct.				the score by one.	
OQ-F-008	The system shall keep the score unchanged for an incorrect answer.	Functional	High	User	An incorrect answer does not increase the score.	
OQ-F-009	The system shall move to the next question after an answer is processed.	Functional	High	User	The next question is displayed after a valid answer.	
OQ-F-010	The system shall calculate the final score after all questions are answered.	Functional	High	User	Final score equals the number of correct answers.	
OQ-F-011	The system shall display the final score to the user.	Functional	High	User	The final score is clearly displayed after the quiz.	
OQ-F-012	The system shall allow the administrator to add a question.	Functional	Medium	Admin	A valid new question is stored and can be displayed.	

OQ-F-013	The system shall allow the administrator to modify an existing question.	Functional	Medium	Admin	The updated question is saved correctly.	
OQ-F-014	The system shall allow the administrator to delete an existing question.	Functional	Medium	Admin	The selected question is removed from the quiz data.	
OQ-F-015	The system shall prevent unauthenticated users from accessing question-management functions.	Functional	High	Security	Unauthorized access is denied.	

4.2.Question Management

Description: Define administrator functions for adding, modifying, and deleting quiz questions.

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
OQ-F-016	The system shall allow the user to restart the quiz after viewing the result.	Functional	Medium	User	A new quiz starts with the score reset.	

4.3.User and Access Control

Requirements: The system shall control access to administrator functions and prevent unauthorized question management.

5. Non-functional requirements (detailed)

NFRs below are measurable and tied to test plans. IDs follow OQ-NF-###.

Req ID	Requirement	Category	Priority	Acceptance criteria / Measurement
OQ-NF-001	The system should respond to normal user input within 1 second.	Performance	Medium	95% of inputs are processed within 1 second.
OQ-NF-002	The system should correctly preserve the score throughout a quiz.	Reliability	High	Repeated runs with the same answers produce the same score.
OQ-NF-003	The console messages and instructions should be clear and easy to understand.	Usability	Medium	A new user can follow the quiz without extra instructions.
OQ-NF-004	Administrator question data should not be exposed to unauthorized users through the normal user menu.	Security	High	Unauthorized access to question management is denied.
OQ-NF-005	The program should run on a standard	Portability	Medium	The application builds and runs in the

	machine with a supported C/C++ compiler without special hardware.			supported environment.

1. Security

1.1. Security Objectives

Security objectives:
1. Prevent unauthorized changes to quiz questions and answers.
2. Protect administrator access and keep credentials out of normal output.

1.2. Security Requirements

Req ID	Requirement (shall...)	Type	Priority	Acceptance criteria / Test case ref
OQ-SR-001	Administrator login shall be required before question-management functions can be used.	Security	High	Login is required before add, modify, or delete operations.
OQ-SR-002	The system shall reject incorrect administrator credentials.	Security	High	Incorrect credentials do not provide access.
OQ-SR-003	The system shall validate administrator input before saving question data.	Security	Medium	Incomplete or invalid question data is rejected.
OQ-SR-004	The system shall prevent normal users from opening the administrator menu directly.	Security	High	Direct unauthorized access is denied.
OQ-SR-005	The system shall avoid displaying administrator credentials in normal console output.	Security	Medium	Credentials are not printed back to the screen.

6. Quality attributes & Acceptance tests

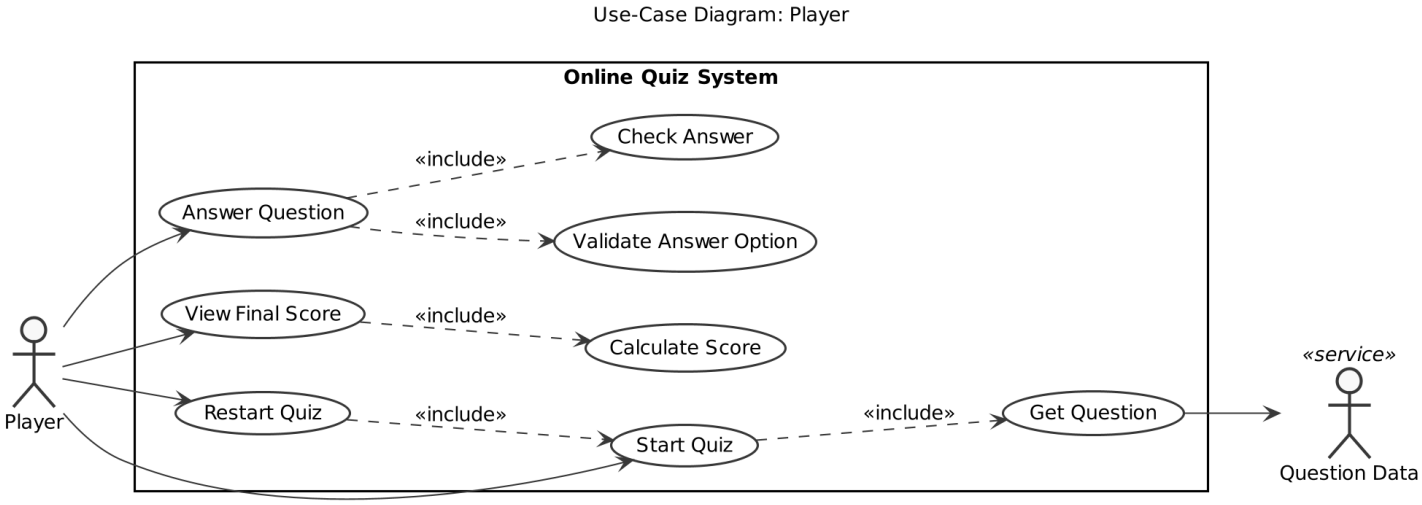
- Exit criteria: all high-priority functional requirements are implemented and verified, no critical security issue remains, and the RTM is updated.
- Acceptance checks: quiz flow, answer validation, scoring, question management, invalid input handling, and administrator access.
- Acceptance test suites: Authentication, Withdrawal, Deposit, Transfer, Performance, Security, and Accessibility tests.

7. System models and diagrams

7.1.UML Use-Case diagram

The following use-case diagram shows the main interaction between the User, Administrator, and Online Quiz System.

Figure 1. UML Use-Case Diagram – Quiz System



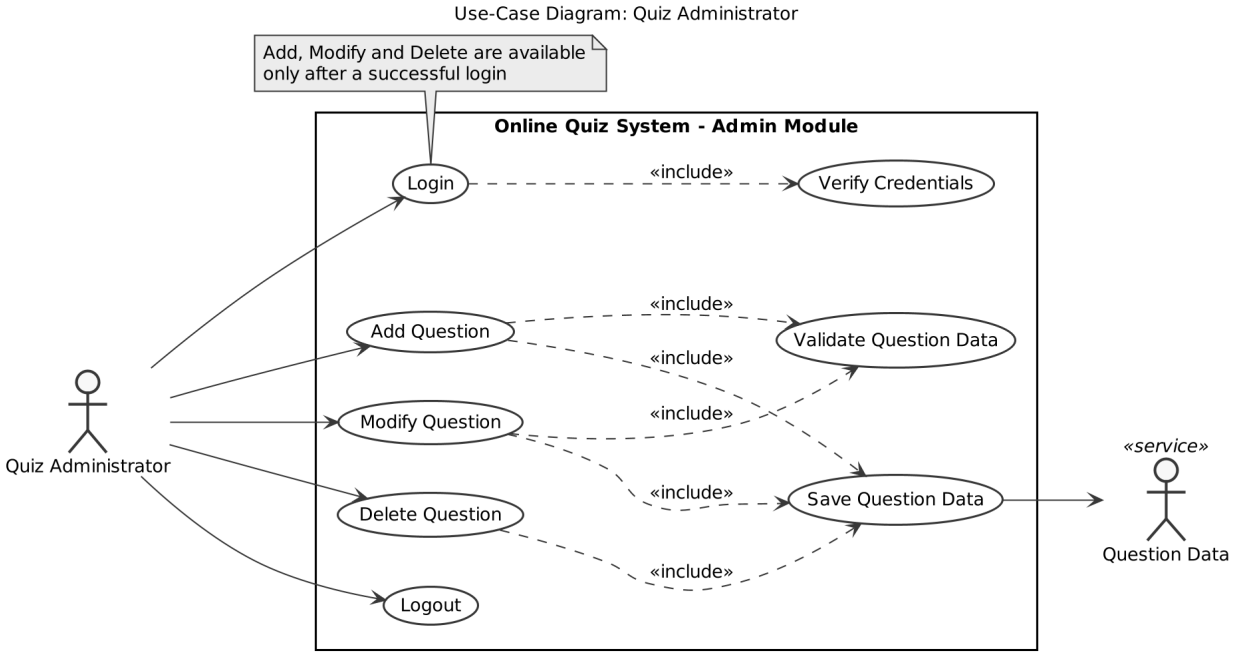


Figure 2. UML Use-Case Diagram – Admin Module

8. Requirements Traceability Matrix (RTM)

Req ID	Requirement short	Section ref / Design Spec	Module	Test case(s)	Status (N/P/A)	Comments
OQ-F-001	The system shall start a new quiz when the user sele...	4 / Design	Quiz Controller / Quiz Logic	TC-01	N	

OQ-F-002	The system shall display one multiple-choice questio...	4 / Design	Quiz Controller / Quiz Logic	TC-02	N	
OQ-F-003	The system shall display four answer options for eac...	4 / Design	Quiz Controller / Quiz Logic	TC-03	N	
OQ-F-004	The system shall accept the user's selected answer.	4 / Design	Quiz Controller / Quiz Logic	TC-04	N	
OQ-F-005	The system shall reject an invalid answer option and...	4 / Design	Quiz Controller / Quiz Logic	TC-05	N	

OQ-F-006	The system shall compare the selected answer with th...	4 / Design	Quiz Controller / Quiz Logic	TC-06	N	

OQ-F-007	The system shall increase the score when the selecte...	4 / Design	Quiz Controller / Quiz Logic	TC-07	N	
OQ-F-008	The system shall keep the score unchanged for an inc...	4 / Design	Quiz Controller / Quiz Logic	TC-08	N	
OQ-F-009	The system shall move to the next question after an...	4 / Design	Quiz Controller / Quiz Logic	TC-09	N	
OQ-F-010	The system shall calculate the final score after all...	4 / Design	Quiz Controller / Quiz Logic	TC-10	N	

OQ-F-011	The system shall display the final score to the user.	4 / Design	Quiz Controller / Quiz Logic	TC-11	N	
OQ-F-012	The system shall allow the	4 / Design	Admin Module	TC-12	N	

	administrator to add a qu...					
OQ-F-013	The system shall allow the administrator to modify a...	4 / Design	Admin Module	TC-13	N	
OQ-F-014	The system shall allow the administrator to delete a...	4 / Design	Admin Module	TC-14	N	
OQ-F-015	The system shall prevent unauthenticated users from...	4 / Design	Admin Module	TC-15	N	

OQ-F-016	The system shall allow the user to restart the quiz...	4 / Design	Quiz Controller / Quiz Logic	TC-16	N	
OQ-NF-001	The system should respond to normal user input withi...	5 / Design	Console UI / Core Logic	TC-17	N	

OQ-NF-002	The system should correctly preserve the score throu...	5 / Design	Console UI / Core Logic	TC-18	N	
OQ-NF-003	The console messages and instructio ns should be clea...	5 / Design	Console UI / Core Logic	TC-19	N	
OQ-NF-004	Administrator question data should not be exposed to...	5 / Design	Console UI / Core Logic	TC-20	N	

OQ-NF-005	The program should run on a standard machine with a...	5 / Design	Console UI / Core Logic	TC-21	N	
OQ-SR-001	Administrator login shall be required before questio...	5.1 / Security Design	Admin Module	TC-22	N	
OQ-SR-002	The system shall reject incorrect administrator cred...	5.1 / Security Design	Admin Module	TC-23	N	
OQ-SR-003	The system shall validate administrator input before...	5.1 / Security Design	Admin Module	TC-24	N	

OQ-SR-004	The system shall prevent normal users from opening t...	5.1 / Security Design	Admin Module	TC-25	N	
OQ-SR-005	The system shall avoid displaying administrator cred...	5.1 / Security Design	Admin Module	TC-26	N	